International Baccalaureate Internal Assessment Mathematics Analysis and Approaches Higher Level

Exploration topic: Comparing forecasted call option prices with market data from Tehran Stock Exchange using the Black-Scholes model.

Aim: To Determine if there is a significant gap between the model's valuation of a call option and its market valuation.

Candidate code: klb698

Table of Contents

1: Introduction

1.1: Rationale

Since I can remember, my father was a stock market enthusiast. When I was 12, I got interested in trading and so my father taught me the fundamentals of trading and what technologies and tools I can use to my benefit. Being in a highly volatile and unpredictable market like Iran's, I was told that a lot of the useful techniques that work for developed stock market's don't translate well onto the Tehran Stock Exchange. One famous example would be the Black-Scholes Model, which is used to determine a fair price for a European options contract. The Black-Scholes model worked great till the 2008 market crash, until option pricing wasn't as stable anymore and had more variables to take into account. I wanted to put my father's claim to the test, and see how effective a foreign model would be if applied to the Tehran stock exchange, which has led me to my research question "***Do forecasted call option prices using the Black-Scholes model significantly differ with market call option data from Tehran Stock Exchange?"***

1.2: Aim

My aim was to be able to carefully craft a Black-Scholes model and compare it to a model of the actual values of IKCO (Iran Khodro) gathered from the Tehran stock exchange over a 5-month period. To do so accurately, I will be calculating the log returns of the models. I will then compare the accuracy of the Black-Scholes model's values by calculating the difference between predicted and the actual values of the call options using RMSE (Root Mean Square Error). In order to find out substantial difference between the model values and the Tehran stock exchange, I have formulated a Null hypothesis.

Null Hypothesis (HO): *There is no significant difference between Black-Scholes values and the Actual market values.*

Alternate Hypothesis (H1): *There is significant difference between Black-Scholes values and the Actual market values.*

1.3: Background Information

With regard to finance, an option can be described as a contract in which the seller promises that the buyer has the right, but not the obligation, to buy or sell a security at a certain price up until, or at, its expiration date. European-style options contracts can only be exercised when the contract reaches its expiration date. The Tehran stock exchange is based off European-style options. The Black-Scholes Formula is one of the most well-known models for calculating theoretical European option prices. Fischer Black, Myron Scholes, and Robert Merton, three economists, first presented the model in a 1973 paper titled "The Pricing of Options and Corporate Liabilities," which was published in the Journal of Political Economy (Hayes, 2022).

The Black-Scholes Model is based off an Ideal market with no arbitrage and full of transactions. This doesn't match well with the Iranian market since there is not a lot of call option training on a daily basis in Iran, and arbitrage is seen often. Since the market direction can never be accurately predicted, a state of random walk—a process for determining the likely location of a point subject to random motions, given the probability weighting—is assumed in the model to match the highly volatile Iranian market. In the Black-Scholes model, no transaction costs, such as commission and brokerage, are assumed. However, there are typically transaction costs associated with trading, such as brokerage fees, commissions, etc. The Black-Scholes model makes the supposition that the stocks don't generate any returns or dividends. However, this is not true in the real trading market. The underlying asset is assumed to be risk-free because it is assumed that interest rates will remain constant, but this is rarely the case in Iran because interest rates are constantly fluctuating. The Black-Scholes-Merton model relies on the principle that asset prices are bounded by zero and have a lognormal distribution, which prevents asset prices from going negative (Trinidad, 2022.) For calculating and modeling the fair price of a call option using the Black-Scholes model, I will use Microsoft Excel 2013, as it is the number one platform when it comes to analyzing and graphing data. It also has multiple built-in mathematical formulas and methods that can be used for calculations.

2: Finding the Black-Scholes Equation

2.1: Introductions to terms and data

The fair price of a call option cannot be determined with certainty because it depends on the future value of the underlying stock, which is unknown. The stock's price is based off Brownian motion, meaning there is randomness in the price, causing high risk. The formula for the change in stock in denoted by (Shreve, 2004):

$$dS_{\mathrm{t}} = \mu S_{\mathrm{t}}\mathrm{d}t + \sigma S_{\mathrm{t}}\mathrm{d}W_t \tag{1}$$

with,

$$S_{\mathrm{t}} = \text{stock price at time t}, \mu = \text{drift}, \sigma = \text{volatility}, W_t = \text{weiner process}$$

The first term is the drift/growth rate term and the second term is the volatility/uncertainty term, $\mathrm{d}W_t$ term is causing the risk in the stock price. In order to find the price of the call option, we need to make it risk-free. We can do so by setting our self-financing portfolio (a collection of assets an individual owns, typically including stocks, bonds and cash.) This portfolio will be comprised of an option, Δ units of the underlying stock, and a price of the stock, S. The goal is to make our portfolio risk-free, i.e. insensitive to changes in the price of the security. This method is called delta hedging.

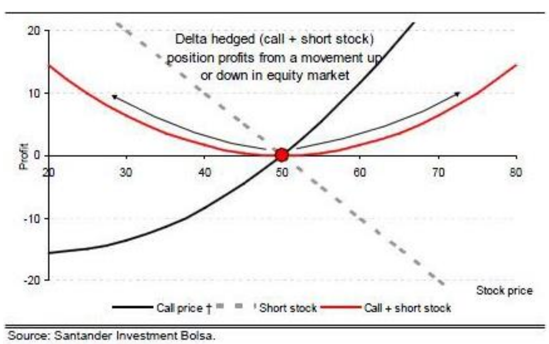


Figure 1 - Delta Hedging Call (Volatility Engineering and Volatility Trading, 2013)

We will buy a call option, and sell Δ stocks against it make the portfolio risk free, as it can be seen in **figure 1**.

The formula for a portfolio can be written as:

$$\mathbf{\Pi} = V(S,t) - \Delta S_{\mathrm{t}} \tag{2}$$

with,

$$\mathbf{\Pi} = \text{portofolio}, S = \text{stock price}, t = \text{time to maturity}, \Delta = \text{number od stocks}$$

2.2: Deriving for the Black-Scholes PDE using Delta Hedging

We are interested in how the portfolio evolves through time, since if there are risk factors in the evolution, we want to cancel them out. So we will that the derivative of our portfolio (Perfiliev Financial Training, 2021):

$$d\mathbf{\Pi} = dV(S,t) - \Delta dS_{\mathrm{t}} \tag{3}$$

To figure out dV, we'll have to use the partial Taylor series:

$$dV = \frac{\partial V}{\partial t}dt + \frac{\partial V}{\partial S}dS_{\mathrm{t}} + \frac{1}{2}\left(\frac{\partial^2 V}{\partial S^2}(dt)^2 + \frac{\partial^2 V}{\partial S^2}(dS_{\mathrm{t}})^2\right) + \frac{1}{6}\left(\frac{\partial^3 V}{\partial S^3}(dt)^3 + \frac{\partial^3 V}{\partial S^3}(dS_{\mathrm{t}})^3\right) \dots$$

Geometric Brownian Motion follow these rules as $dt \to 0$

$$(\mathrm{dt})^k = 0, K > 1 \tag{4}$$

$$(\mathrm{d}W_t)^k = 0, K > 2 \tag{5}$$

$$dtdW_t = 0 \tag{6}$$

And due to the Weiner process,

$$dW_t = N(0,1)\sqrt{dt} \tag{7}$$

with $N(0,1)$ being the normal distribution with a mean of zero and standard deviation of one.

Due to these boundaries, the only dt term that will remain is the first order $\frac{\partial V}{\partial t}dt$ term, as per (4). We also know the value of dS from before (1), so to plugging replacing the value of dS to find the value of dS^2, we do a binomial expansion:

$$(dS_{\mathrm{t}})^2 = (\mu S_{\mathrm{t}}\mathrm{c} + \sigma S_{\mathrm{t}}\mathrm{dW})^2 = \mu^2 {S_{\mathrm{t}}}^2(dt)^2 + 2\sigma\mu {S_{\mathrm{t}}}^2 dtdW + \sigma^2 {S_{\mathrm{t}}}^2(dW)^2 = \sigma^2 {S_{\mathrm{t}}}^2(dt)^2$$

Due to the rules of Geometric Brownian Motion, the 1st and 2nd terms of the expansion cancel, and the $(dW)^2$ becomes dt. $(\mathrm{dS})^k = 0, K > 2$, since $(\mathrm{dW})^k = 0, K > 2$. So the only terms left in the Taylor series is the first order dt term, and the first & second order dS terms. Simplifying the Taylor series:

$$dV = \frac{\partial V}{\partial t}dt + \frac{\partial V}{\partial S}dS_{\mathrm{t}} + \frac{1}{2}\frac{\partial^2 V}{\partial S^2}(dS_{\mathrm{t}})^2$$

Replace the dS^2 term with its solution and let dS be as it is.

$$dV = \frac{\partial V}{\partial t}dt + \frac{\partial V}{\partial S}dS_t + \frac{1}{2}\sigma^2 {S_t}^2 \frac{\partial^2 V}{\partial S^2}\ dt$$

Now that we have dV, we can group the dt terms together and plug dV back into our change in portfolio (2):

$$d\mathbf{\Pi} = dV(S,t) - \Delta dS_t$$

$$d\mathbf{\Pi} = \frac{\partial V}{\partial t}dt + \frac{\partial V}{\partial S}dS_t + \frac{1}{2}\sigma^2 {S_t}^2 \frac{\partial^2 V}{\partial S^2}\ dt + \Delta dS_t$$

$$d\mathbf{\Pi} = \left[\frac{\partial V}{\partial t} + \frac{1}{2}\sigma^2 {S_t}^2 \frac{\partial^2 V}{\partial S^2}\right]dt + \left|\frac{\partial V}{\partial S} - \Delta\right| dS_t$$

We can set delta to $\frac{\partial V}{\partial S}$ to cancel out dS, canceling the randomness leaving only a dt term, which makes the portfolio deterministic, hence risk-free. Since the portfolio is risk-free, we can use a risk-free rate r. This could be the rate of return of a bank, for example. We can denote investment in a bank as $B_t = rB_t dt$. Therefore,

$$\mathbf{\Pi} = B_t$$

$$d\mathbf{\Pi} = r\mathbf{\Pi}dt$$

We can substitute $d\mathbf{\Pi}$ and $\mathbf{\Pi}$ for their equations and get:

$$\left[\frac{\partial V}{\partial t} + \frac{1}{2}\sigma^2 {S_t}^2 \frac{\partial^2 V}{\partial S^2}\right]dt = \left[rV - rS\frac{\partial V}{\partial S}\right]dt$$

Cancel dt terms on both sides of the equation. Make the equation equal to zero to get the Black-Scholes PDE:

$$\frac{\partial V}{\partial t} + \frac{1}{2}\sigma^2 {S_t}^2 \frac{\partial^2 V}{\partial S^2} + rS\frac{\partial V}{\partial S} - rV = 0 \tag{8}$$

With Boundary conditions (Chello, 2022):

$$V(S,t) = (S-K)^+ = \max(0, S-K) \tag{9}$$

2.3: Converting the Black-Scholes PDE into the Diffusion Equation

Now we have a PDE for the price of a call option, but it is not solvable for the price. If we apply a few transformations and simplify the equation to get the diffusion equation, the solution to which we know, then we can apply the formula to get the price of a call option.

It can be said that the PDE is a backward equation, since we are considering the stock price to be $S(T)$, the price at maturity and we are discounting it to get $S(t)$, the stock price today (quantpie, 2019) . We can make the PDE forward by considering time $\tau = T - t$, making $S(T) = \tilde{S}(0)$ and $S(t) = \tilde{S}(\tau)$. In Geometric Brownian Motion (Krouglov, n.d.),

$$S(T) = e^{\ln S + (r - \frac{1}{2}\sigma^2)(T-t)} \text{, So } \tilde{S}(0) = e^{\ln \tilde{S} + (r - \frac{1}{2}\sigma^2)(\tau)}$$

To simplify the equation take ln from sides, and name it x,

$$x = \ln \tilde{S}(0) = \tilde{S} + (r - \frac{1}{2}\sigma^2)(\tau)$$

And finally, we transform the current value of the option price to its forward value.

$$F(\tau) = \tilde{V} e^{r\tau}$$

Now that we have our formulas, let's start by reversing the time. Let's first derive τ, with respect to t:

$$\tau = T - t \text{, So } \frac{\partial \tau}{\partial t} = -1$$

We need to transform the $\frac{\partial V}{\partial t}$ to $\frac{\partial V}{\partial \tau}$ using the chain rule:

$$\frac{\partial V}{\partial t} = \frac{\partial \tilde{V}}{\partial \tau} \frac{\partial \tau}{\partial t} = -\frac{\partial \tilde{V}}{\partial \tau}$$

We plug $-\frac{\partial \tilde{V}}{\partial \tau}$ back into the PDE:

$$-\frac{\partial \tilde{V}}{\partial \tau} + \frac{1}{2}\sigma^2 S_t^2 \frac{\partial^2 V}{\partial S^2} + rS_t \frac{\partial V}{\partial S} - rV = 0$$

Now we need to transform stock price, first derive x with respect to $\tilde{S}$:

$$x = \ln \tilde{S}(0) = \tilde{S} + (r - \frac{1}{2}\sigma^2)(\tau) \text{, So } \frac{\partial x}{\partial \tilde{S}} = \frac{1}{\tilde{S}}$$

We need to transform $\frac{\partial \tilde{V}}{\partial \tilde{S}}$ to $\frac{\partial V}{\partial \tilde{x}}$ using the chain rule:

$$\frac{\partial \tilde{V}}{\partial \tilde{S}} = \frac{\partial v}{\partial \mathrm{x}} \frac{\partial \mathrm{x}}{\partial \tilde{S}} = \frac{\partial v}{\partial \mathrm{x}} \frac{1}{\tilde{S}}$$

We also need to take the 2nd partial derivative of v with respect to x using the product and chain rule:

$$\frac{\partial^2 v}{\partial \tilde{S}^2} = \frac{\partial}{\partial \tilde{S}}\left(\frac{\partial v}{\partial \mathrm{x}}\frac{1}{\tilde{S}}\right) = -\frac{\partial v}{\partial \mathrm{x}}\frac{1}{\tilde{S}^2} + \frac{1}{\tilde{S}}\frac{\partial}{\partial \tilde{S}}\left(\frac{\partial v}{\partial \mathrm{x}}\right) = -\frac{\partial v}{\partial \mathrm{x}}\frac{1}{\tilde{S}^2} + \frac{1}{\tilde{S}}\frac{\partial}{\partial \tilde{S}}\left(\frac{\partial v}{\partial \mathrm{x}}\right)\frac{\partial x}{\partial \tilde{S}} =$$

$$-\frac{\partial v}{\partial \mathrm{x}}\frac{1}{\tilde{S}^2} + \frac{1}{\tilde{S}^2}\frac{\partial^2 v}{\partial x^2} = \frac{1}{\tilde{S}^2}\left(-\frac{\partial v}{\partial \mathrm{x}} + \frac{\partial^2 v}{\partial x^2}\right)$$

We also need to transform $\frac{\partial \tilde{V}}{\partial \tau}$, so derive x with respect to τ:

$$x = \ln \tilde{S}(0) = \tilde{S} + (r - \tfrac{1}{2}\sigma^2)(\tau), \text{ So } \frac{\partial x}{\partial \tau} = r - \tfrac{1}{2}\sigma^2$$

A transformation of will also produce a total derivate:

$$\frac{\partial \tilde{V}}{\partial \tau} = \frac{\partial v}{\partial \tau} + \frac{\partial v}{\partial \mathrm{x}}\frac{\partial x}{\partial \tau} = \frac{\partial v}{\partial \tau} + \frac{\partial v}{\partial \mathrm{x}}\left(r - \frac{1}{2}\sigma^2\right)$$

Plugging these three back into the PDE, canceling the S_t and ${S_t}^2$ terms:

$$-\frac{\partial v}{\partial \tau} - \frac{\partial v}{\partial \mathrm{x}}\left(r - \frac{1}{2}\sigma^2\right) + \frac{1}{2}\sigma^2\left(-\frac{\partial v}{\partial \mathrm{x}} + \frac{\partial^2 v}{\partial x^2}\right) + r\frac{\partial v}{\partial x} - rv = 0$$

Now we rearrange the 3rd and 4th terms to isolate the 1st and 2nd derivate terms:

$$-\frac{\partial v}{\partial \tau} - \frac{\partial v}{\partial \mathrm{x}}\left(r - \frac{1}{2}\sigma^2\right) + \frac{1}{2}\sigma^2\frac{\partial^2 v}{\partial x^2} + \frac{\partial v}{\partial x}\left(r - \frac{1}{2}\sigma^2\right) - rv = 0$$

The 2nd and the penultimate terms cancel:

$$-\frac{\partial v}{\partial \tau} + \frac{1}{2}\sigma^2\frac{\partial^2 v}{\partial x^2} - rv = 0$$

We move on to the Final transformation:

$$F(\tau) = \tilde{V}e^{r\tau} \rightarrow v = Fe^{-r\tau}$$

We inverted the relationship to make the comparison with the PDE clearer. We need the second derivatives with respect to x:

$$\frac{\partial v}{\partial \mathrm{x}} = e^{-r\tau}\frac{\partial F}{\partial \mathrm{x}} \text{ Therefore, } \frac{\partial^2 v}{\partial x^2} = e^{-r\tau}\frac{\partial^2 F}{\partial x^2}$$

We also need to take the derivative with respect to τ using the product rule:

$$\frac{\partial v}{\partial \tau} = e^{-r\tau}\frac{\partial F}{\partial \tau} + F\frac{\partial}{\partial \tau}e^{-r\tau} = e^{-r\tau}\frac{\partial F}{\partial \tau} - rFe^{-r\tau}$$

Plugging the values back into the PDE, we get:

$$-e^{-r\tau}\frac{\partial F}{\partial \tau} - rFe^{-r\tau} + \frac{1}{2}\sigma^2 e^{-r\tau}\frac{\partial^2 F}{\partial x^2} + rFe^{-r\tau} = 0$$

The 2nd and last term cancel out to get:

$$-e^{-r\tau}\frac{\partial F}{\partial \tau} + \frac{1}{2}\sigma^2 e^{-r\tau}\frac{\partial^2 F}{\partial x^2} = 0$$

Taking the 2nd and 3rd term to the RHS, and dividing both sides by $-e^{-r\tau}$, we get the heat/diffusion equation:

$$\frac{\partial F}{\partial \tau} = \frac{1}{2}\sigma^2 \frac{\partial^2 F}{\partial x^2} \tag{10}$$

Finally, we need to change the Boundary and Terminal conditions. Since we reversed the equation:

$$V(T) = \max(0, S - K) \rightarrow \tilde{V} = max(\tilde{S}_0 - K, 0)$$

$$\tilde{S}(\tau) = e^{x + (r - \frac{1}{2}\sigma^2)(\tau)} \rightarrow \tilde{S}(0) = e^x$$

$$\tilde{V} = F(\tau)e^{r\tau} \rightarrow \tilde{V} = F(0)$$

$$F(x, 0) = max(e^x - K, 0)$$

2.4: Simplifying the equation to get the Black-Scholes Equation

In stochastic calculus, the heat equation is referred to as the diffusion equation.

If we have a diffusion equation $\frac{\partial f}{\partial \tau} = D\frac{\partial^2 f}{\partial x^2}$ subject to initial conditions $F(x, 0) = \Psi(x)$, then we know it's **fundamental solution in 1 dimension** is as follows (quantpie, 2019b):

$$F(x, t) = \int_{-\infty}^{\infty} \Psi(z)\, \frac{1}{\sqrt{4\pi D t}}\, e^{-\frac{(x-z)^2}{4Dt}}\, dz \tag{11}$$

In our PDE, $D = \frac{1}{2}\sigma^2$, so our solution is:

$$F(x, \tau) = \int_{-\infty}^{\infty} (e^z - K)^+ \left(\frac{1}{\sqrt{4\pi \frac{1}{2}\sigma^2\tau}}\, e^{-\frac{(x-z)^2}{4\frac{1}{2}\sigma^2\tau}}\, dz = \int_{-\infty}^{\infty} (e^z - K)^+ \,\frac{1}{\sigma\sqrt{2\pi\tau}}\, e^{-\frac{(x-z)^2}{2\sigma^2\tau}}\, dz\right.$$

$(e^z - K)^+$ is positive since the option won't be exercised if stock price was lower than strike price. The expression is non-negative, only when:

$$e^z - K \geq 0 \rightarrow e^z \geq K$$

We can isolate z by taking ln of both sides: $z \geq \ln K$, meaning the integral only counts when z is greater than $\ln K$.

$$\int_{\ln K}^{\infty} (e^z - K) \frac{1}{\sigma\sqrt{2\pi\tau}} e^{-\frac{(x-z)^2}{2\sigma^2 t}} dz$$

We expand the equation:

$$\int_{\ln K}^{\infty} e^z \frac{1}{\sigma\sqrt{2\pi\tau}} e^{-\frac{(x-z)^2}{2\sigma^2\tau}} dz - K \int_{\ln K}^{\infty} \frac{1}{\sigma\sqrt{2\pi\tau}} e^{-\frac{(x-z)^2}{2\sigma^2\tau}} dz$$

We will first solve the 2nd term to get into the swing of things. We see that the integrand is essentially the probability density of a normal:

$$z \sim N[x, \sigma\sqrt{\tau}]$$

Notice for the $-(x - z) = z - x$. We know how to standardize a normal based of probability (Hease et al., 2019):

$$Z = \frac{z - x}{\sigma\sqrt{\tau}} \sim N[0,1]$$

We can rearrange to give in terms of z,

$$z = x + \sigma\sqrt{\tau}Z$$

This so the integral essentially represents the probability of

$$\text{Prob}(z \geq \ln K)$$

We substitute in the expression for z, then isolate Z on the LHS:

$$\text{Prob}(x + \sigma\sqrt{\tau}Z \geq \ln K) \rightarrow \text{Prob}\left(Z \geq \frac{\ln K - x}{\sigma\sqrt{\tau}}\right)$$

the standard normal is symmetric about 0, so

$$\text{Prob}\left(Z \geq \frac{\ln K - x}{\sigma\sqrt{\tau}}\right) \rightarrow \text{ Prob}\left(Z \leq -\frac{\ln K - x}{\sigma\sqrt{\tau}}\right)$$

if represent the distribution of the standard normal by Φ,

$$\Phi\left(-\frac{\ln K - x}{\sigma\sqrt{\tau}}\right)$$

This is equivalent to the integral the term in out $F(x,\tau)$ equation, so multiplying it by $-K$, we get:

$$\int_{\ln K}^{\infty} e^{z} \frac{1}{\sigma\sqrt{2\pi\tau}}\, e^{-\frac{(x-z)^2}{2\sigma^2\tau}}\, dz - K\,\Phi\left(-\frac{\ln K - x}{\sigma\sqrt{\tau}}\right)$$

Now to move on the getting the solution of the first term, first we expand the binomial:

$$\int_{\ln K}^{\infty} \frac{1}{\sigma\sqrt{2\pi\tau}}\, e^{z} e^{-\frac{x^2 z^2 - 2xz}{2\sigma^2\tau}} dz$$

Afterwards, we can combine the two exponentials:

$$\int_{\ln K}^{\infty} \frac{1}{\sigma\sqrt{2\pi\tau}}\, e^{-\frac{-2z\sigma^2\tau + x^2 z^2 - 2xz}{2\sigma^2\tau}} dz$$

We can split out x, since it does not depends on the integrating variable:

$$\int_{\ln K}^{\infty} \frac{1}{\sigma\sqrt{2\pi\tau}}\, e^{-\frac{x^2}{2\sigma^2\tau}} e^{-\frac{z^2 - 2xz - 2z\sigma^2\tau}{2\sigma^2\tau}} dz$$

We need to simplify the numerator of the 2nd exponential by completing the square:

$$z^2 - 2xz - 2z\sigma^2\tau = \; z^2 - 2z(x + \sigma^2\tau) + (x + \sigma^2\tau)^2$$

Since we added $(x + \sigma^2\tau)^2$ to the second exponential, we need to subtract it from the first exponential:

$$\int_{\ln K}^{\infty} \frac{1}{\sigma\sqrt{2\pi\tau}}\, e^{-\frac{x^2 - (x+\sigma^2\tau)^2}{2\sigma^2\tau}}\, e^{-\frac{z^2 - 2z(x+\sigma^2\tau) + (x+\sigma^2\tau)^2}{2\sigma^2\tau}} dz$$

We expand the first exponential and simplify it, while we get a complete square in the second exponential.

$$\int_{\ln K}^{\infty} \frac{1}{\sigma\sqrt{2\pi\tau}}\, e^{x - \frac{\sigma^2\tau}{2\sigma^2\tau}}\, e^{-\frac{\left(z - (x+\sigma^2\tau)\right)^2}{2\sigma^2\tau}}\, dz$$

As the terms in the first exponential don't depend on the Integrating variable, we can take it out:

$$e^{x-\frac{\sigma^2\tau}{2\sigma^2\tau}}\int_{\ln K}^{\infty}\frac{1}{\sigma\sqrt{2\pi\tau}}\, e^{-\frac{\left(z-(x+\sigma^2\tau)\right)^2}{2\sigma^2\tau}}\, dz$$

now the integrand is nothing but the density of a normal:

$$z \sim N[x+\sigma^2\tau, \sigma^2\tau]$$

From earlier on, we know how to write this in terms of the standard normal distribution:

$$e^{x-\frac{\sigma^2\tau}{2\sigma^2\tau}}\int_{\ln K}^{\infty}\frac{1}{\sigma\sqrt{2\pi\tau}}\, e^{-\frac{\left(z-(x+\sigma^2\tau)\right)^2}{2\sigma^2\tau}}\, dz = \; e^{x-\frac{\sigma^2\tau}{2\sigma^2\tau}\Phi}\left(-\frac{\ln K + x + \sigma^2\tau}{\sigma\sqrt{\tau}}\right)$$

Therefore, we find the solution to the heat equation:

$$e^{x-\frac{\sigma^2\tau}{2\sigma^2\tau}\Phi}\left(-\frac{\ln K + x + \sigma^2\tau}{\sigma\sqrt{\tau}}\right) - K\Phi\left(-\frac{\ln K - x}{\sigma\sqrt{\tau}}\right)$$

Now we need to reverse the changes we made initially to make it a backward PDE

$$\tau \;=\; \mathrm{T} - \mathrm{t}$$

$$x = \ln\tilde{S}(0) = \tilde{S} + (r - \frac{1}{2}\sigma^2)(\tau)$$

$$F(\tau) = \tilde{V}e^{r\tau}$$

First, we set the function equal to $\tilde{V}e^{r\tau}$:

$$\tilde{V}e^{r\tau} = RHS$$

Then, we substitute the solution of x into the equation:

$$\tilde{V}e^{r\tau} = \; e^{\ln\tilde{S}+\left(r-\frac{1}{2}\sigma^2\right)(\tau)+\frac{\sigma^2\tau}{2}\Phi}\left(-\frac{\ln K + \ln\tilde{S} + \left(r+\frac{1}{2}\sigma^2\right)(\tau) + \sigma^2\tau}{\sigma\sqrt{\tau}}\right)$$

$$-K\Phi\left(-\frac{-\ln K - \ln\tilde{S} + (r - \frac{1}{2}\sigma^2)(\tau)}{\sigma\sqrt{\tau}}\right)$$

Simplifying the sigma terms, we get:

$$\tilde{V}e^{r\tau} = e^{\ln\tilde{S}+r\tau}\Phi\left(-\frac{\ln K + \ln\tilde{S} + \left(r + \frac{1}{2}\sigma^2\right)(\tau)}{\sigma\sqrt{\tau}}\right) - K\Phi\left(-\frac{\ln K - \ln\tilde{S} + (r - \frac{1}{2}\sigma^2)(\tau)}{\sigma\sqrt{\tau}}\right)$$

Divide both sides by $e^{r\tau}$ and simplify the ln terms, and we get the **Black-Scholes formula**

$$\tilde{V} = \tilde{S}\Phi\left(\frac{\ln\frac{\tilde{S}}{K} + (r + \frac{1}{2}\sigma^2)(\tau)}{\sigma\sqrt{\tau}}\right) - Ke^{-r\tau}\Phi\left(\frac{\ln\frac{\tilde{S}}{K} + \left(r - \frac{1}{2}\sigma^2\right)(\tau)}{\sigma\sqrt{\tau}}\right) \tag{13}$$

2.5: Explaining the Equation

This is the final formula for finding the fair price of a call option,

$$V(S,\tau) = S\Phi(d1) - Ke^{-r\tau}\Phi(d2)$$

With $d1$ being

$$d1 = \frac{1}{\sigma\sqrt{\tau}}\left[\ln\frac{S}{K} + (r + \frac{1}{2}\sigma^2)(\tau)\right]$$

With $d2$ being

$$d2 = \frac{1}{\sigma\sqrt{\tau}}\left[\ln\frac{S}{K} + (r - \frac{1}{2}\sigma^2)(\tau)\right] \quad \text{or} \quad d2 = d1 - \sigma\sqrt{\tau}$$

Term definitions:

$V(S,t)$ = call option

S = Current stock price

K = Strike price

r = risk-free rate

$\tau = T - t$ = time to maturity

σ = volatility of the asset

Φ = CDF of the normal distribution

3. Application of Mathematical Models

I will be using the Black-Scholes models in order to represent the represent the fair price of a call option in a singular function, as well as modelling the available market data. I've chosen a call option from IKCO (Iran-Khodro), at a strike price of 2000 Rials with a first offering at 28th August, 2022, and an expiry date of 25th January, 2023, spanning 5 months with a total of 98 closing prices.

3.1: Modelling the Option Price from the Black-Scholes Model

The first model I will be employing is the Black-Scholes Model. In order to calculate the fair price of a call option, you need to find the volatility of the underlying asset. I collected 1 trading-year worth of stock price data of IKCO and calculated the annual volatility (Appendix 1). The procedure for calculating the annual volatility is:

$$\log return = \ln\left[\frac{S_{date}}{S_{previous\ date}}\right]$$

$$Daily\ Volatility = \ Variance_{\log returns} = \frac{\sum(x_i - \bar{x})^2}{n-1}$$

$$Annualized\ Volatility = \sqrt{Variance_{\log returns}} \times \sqrt{Number\ of\ trading\ days\ per\ year}$$

Using Microsoft Excel to get the value ("Microsoft Excel," n.d.), the daily volatility was 2.86% and annualized volatility of 42.48%, showcasing the high instability of the Iranian market. Now that we have the annualized volatility, we can calculate the fair price of the call option over the 5-month period. For demonstration purposes, I will solve once instance:

$S_0 = 2104$, K = 2000, r = 20%, current date = 8/28/2023, call option expiry date = 1/25/2023,

τ = time to maturity = 1/25/2023 – 8/28/2023 = 0.41, σ = 42.48%

put the data into the Black-Scholes Equation ("Microsoft Excel," n.d.):

$$V(2104,\ \ 0.41) = \ 2104\Phi(d1) - 2000e^{-0.20\times0.41}\Phi(d2)$$

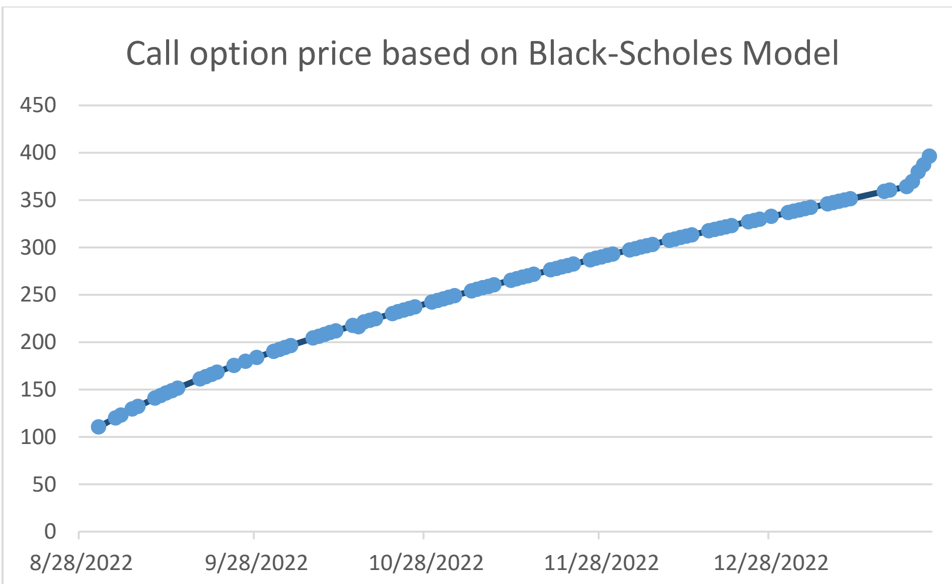
With d1 and d2 being,

$$d1 = \frac{1}{0.4248\sqrt{0.41}}\left[\ln\frac{2104}{2000} + (0.20 + \frac{1}{2}(0.4248)^2)(0.41)\right], d2 = d1 - 0.4248\sqrt{0.42}$$

Plug the values back into the equation:

$$V(2104,\ 0.42) = 2104\Phi(0.6241) - 2000e^{-0.20\times0.41}\Phi(0.3518)\ = \mathbf{369.3532}$$

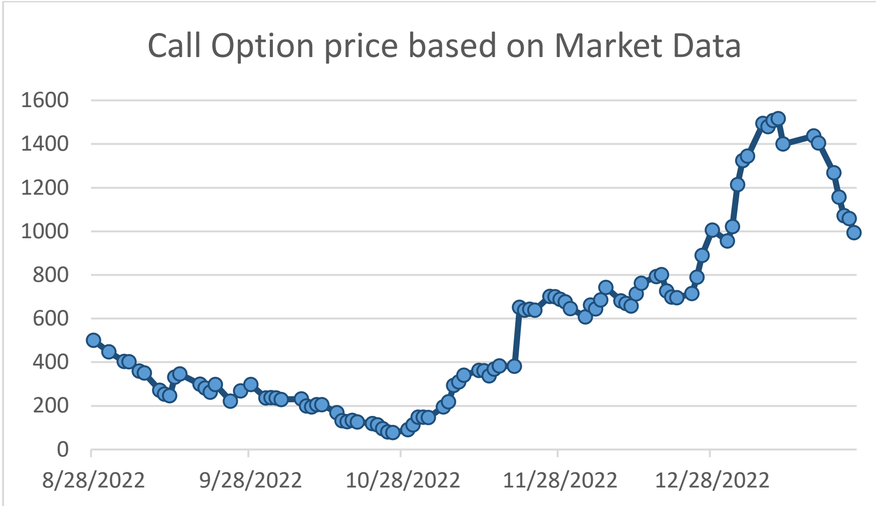
According to the Black-Scholes equation, the fair price of the call option with a maturity of 0.41 is approximately 396 Rials (3 significant figures). Using the same procedure for the rest of the data (Appendix 2) in Microsoft Excel, we can model the theoretical option prices, with the X-axis being the date at which the option was traded, and the Y-axis being the price of the call option.



Graph 1 (Modelled fair price of a call option using the Black-Scholes equation) (Appendix 2) ("Microsoft Excel," n.d.)

3.2: Modelling the Option Price from the Market Data

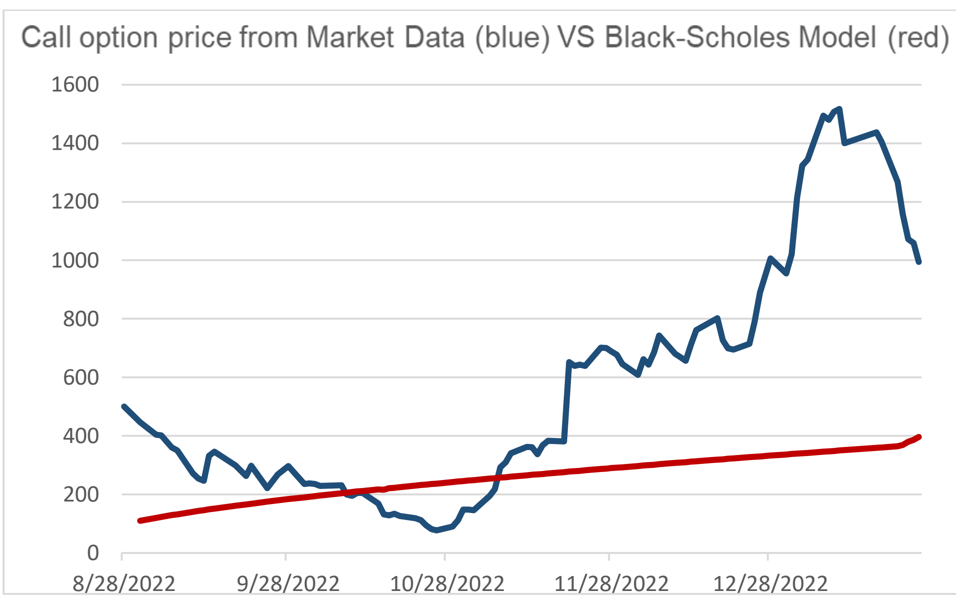
The 2nd model I will be employing is from the real market data released from the Tehran Stock Exchange. The X-axis represents the dates at which the call option was traded, while the Y-axis represents the closing price of the call option on that day. **Graph 2** shows how absurdly volatile the Iranian economy, hence the Tehran Stock Exchange truly is, with the price initially starting at 500 Rials, plummeting to 77 Rials on the 26th of October, peaking at 1517 Rials at the 10th of January, and expiring on 994 Rials.



Graph 2 (Graph of call option price based on market transactions) (Appendix 1) ("Microsoft Excel," n.d.)

4: Comparison of Both Models

Fitting the market data graph (blue) and the Black-Scholes model (red) into a single line graph, we see a substantial difference in the valuation of the call option. What the Black-Scholes model's prediction of the call option price is, is drastically different to the prices being traded in the market.



Graph 3 (Call option price comparison of market data with Black-Scholes model) ("Microsoft Excel," n.d.)

To measure the accuracy of the Black-Scholes Model, I will measure the RMSE (Root-mean-square deviation) of the model. RMSE measures the average difference between values predicted by a model and the actual values. It provides an estimation of how well the model is able to predict the target value (accuracy). The formula and calculation of the RMSE using Microsoft Excel ("Microsoft Excel," n.d.):

$$RMSE = \sqrt{\sum_{i=1}^{n} \frac{(\hat{y}_i - y_i)^2}{n}} = \sqrt{\frac{20{,}588{,}570}{96}} = \mathbf{460.709}$$

The result of the RMSE was approximately 461 Rials (3 significant figures) (Appendix 3), meaning the average closing call option price traded in the market had a 461 price difference with the Black-Scholes' prediction. For a model to be deemed accurate and fit to the data, it has to have an $RMSE \ll 0.5$, proving the Alternate hypothesis that claimed there is significant difference between Black-Scholes values and the Actual market values to be true.

5: Conclusion and Extensions

In this exploration, the fair price of a call option was obtained through the Black-Scholes Model, along with the price of the option traded from the Tehran stock exchange, were compared and evaluated. As the aim of this exploration was to determine whether there is a significant gap between the model's valuation of a call option and its market valuation, I feel the aim has been achieved sufficiently.

I first derived the Black-Scholes Equation using the delta hedging call option, and then used the equation to model the fair price of the call option, on Microsoft Excel 2013, over the maturity of the stock. The model, using RMSE, was compared to the market data for that time period obtained from the Tehran stock exchange. It can be said that although the Black-Scholes Model can accurately model data in more stable economies, it doesn't uphold well in a market with volatility as high as Iran's, proving the alternate hypothesis true. To answer the research question, "***Do forecasted call option prices using the Black-Scholes model significantly differ with market call option data from Tehran Stock Exchange?"*** They do, as there was a significant gap between the Black-Scholes Model and the market data's pricing on the call option. This is likely not a one-time occurrence, as all evidence suggests that the Black-Scholes model does not fit with the Iranian stock market, from the assumption that there are no dividends and arbitrage for the underlying asset to the assumption that the risk-free interest rate is constant, which is rarely the case in a 3rd-world country with high inflation like Iran.

The biggest weakness in my model was that I couldn't gain full access to IKCO's stock data, as the Iranian stock market is more secretive about the company's details. Using a larger set of stock prices for finding the volatility would have made the model more accurate. Additionally, another major limitation of the research was the risk-free rate was taken from the bank's interest rate and not the return of a governmental bond, as no bonds where offered while I was in my researching process. This made the predicted Black-Scholes prices less accurate to fit the market data.

As an extension, I would like to try to model more advanced methods of predicting the theoretical price of a call option, such as Risk-neutral Probability or a Monte-Carlo Simulation, and see if they could work well in the highly volatile Tehran stock exchange, or try the same model but rather put rather for put options and check the model's accuracy and precision.

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Appendix

Appendix I – Volatility of the Underlying Stock from the Year Prior to the Call Option

Date of transaction	Stock Price	Log returns of stock
20220821	2155	
20220820	2171	0.74%
20220817	2220	2.23%
20220816	2151	-3.16%
20220815	2116	-1.64%
20220814	2055	-2.93%
20220813	2008	-2.31%
20220810	1915	-4.74%
20220809	1974	3.03%
20220806	1958	-0.81%
20220803	1871	-4.55%
20220802	1793	-4.26%
20220801	1783	-0.56%
20220731	1770	-0.73%
20220730	1843	4.04%
20220727	1903	3.20%
20220726	1889	-0.74%
20220725	1885	-0.21%
20220724	1982	5.02%
20220723	2003	1.05%
20220716	1998	-0.25%
20220713	2049	2.52%
20220712	2005	-2.17%
20220711	1990	-0.75%
20220709	2064	3.65%
20220706	2078	0.68%
20220705	2126	2.28%
20220704	2151	1.17%
20220703	2112	-1.83%
20220702	2116	0.19%
20220629	2218	4.71%
20220628	2231	0.58%
20220627	2203	-1.26%
20220626	2194	-0.41%
20220625	2202	0.36%
20220622	2100	-4.74%
20220621	2072	-1.34%
20220620	2095	1.10%

20220619	2041	-2.61%
20220615	2081	1.94%
20220614	2137	2.66%
20220613	2113	-1.13%
20220612	2020	-4.50%
20220611	2002	-0.90%
20220608	2041	1.93%
20220607	2025	-0.79%
20220606	2017	-0.40%
20220601	2113	4.65%
20220531	2098	-0.71%
20220530	2116	0.85%
20220529	2110	-0.28%
20220528	2162	2.43%
20220525	2262	4.52%
20220524	2257	-0.22%
20220523	2258	0.04%
20220522	2360	4.42%
20220521	2363	0.13%
20220518	2395	1.35%
20220517	2396	0.04%
20220516	2434	1.57%
20220515	2371	-2.62%
20220514	2358	-0.55%
20220511	2468	4.56%
20220510	2429	-1.59%
20220509	2451	0.90%
20220508	2470	0.77%
20220507	2422	-1.96%
20220502	2420	-0.08%
20220501	2393	-1.12%
20220430	2424	1.29%
20220427	2492	2.77%
20220426	2504	0.48%
20220425	2428	-3.08%
20220424	2441	0.53%
20220420	2429	-0.49%
20220419	2397	-1.33%
20220418	2409	0.50%
20220417	2361	-2.01%
20220416	2428	2.80%
20220413	2396	-1.33%
20220412	2435	1.61%
20220411	2557	4.89%
20220410	2574	0.66%

20220409	2460	-4.53%
20220406	2556	3.83%
20220405	2637	3.12%
20220404	2615	-0.84%
20220403	2588	-1.04%
20220330	2471	-4.63%
20220329	2459	-0.49%
20220328	2554	3.79%
20220327	2506	-1.90%
20220326	2474	-1.29%
20220319	2398	-3.12%
20220316	2284	-4.87%
20220315	2178	-4.75%
20220314	2108	-3.27%
20220313	2082	-1.24%
20220312	2087	0.24%
20220309	2194	5.00%
20220308	2112	-3.81%
20220307	2173	2.85%
20220306	2202	1.33%
20220305	2162	-1.83%
20220302	2061	-4.78%
20220228	2138	3.67%
20220227	2113	-1.18%
20220226	2122	0.43%
20220223	2215	4.29%
20220222	2176	-1.78%
20220221	2167	-0.41%
20220220	2067	-4.72%
20220219	2056	-0.53%
20220216	1990	-3.26%
20220214	2020	1.50%
20220213	1996	-1.20%
20220212	2050	2.67%
20220209	2088	1.84%
20220208	2048	-1.93%
20220207	1974	-3.68%
20220206	2044	3.48%
20220205	2133	4.26%
20220202	2124	-0.42%
20220201	2140	0.75%
20220131	2060	-3.81%
20220130	1997	-3.11%
20220129	1979	-0.91%
20220126	1891	-4.55%

20220125	1816	-4.05%
20220124	1904	4.73%
20220123	1873	-1.64%
20220122	1912	2.06%
20220119	1934	1.14%
20220118	1947	0.67%
20220117	2034	4.37%
20220116	1957	-3.86%
20220115	1957	0.00%
20220112	2054	4.84%
20220111	2069	0.73%
20220110	2002	-3.29%
20220109	1933	-3.51%
20220108	1921	-0.62%
20220105	1833	-4.69%
20220104	1817	-0.88%
20220103	1881	3.46%
20220102	1889	0.42%
20220101	1905	0.84%
20211229	1997	4.72%
20211228	2010	0.65%
20211227	1990	-1.00%
20211226	1905	-4.37%
20211225	1956	2.64%
20211222	1960	0.20%
20211221	1882	-4.06%
20211220	1858	-1.28%
20211219	1876	0.96%
20211218	1953	4.02%
20211215	1868	-4.45%
20211214	1921	2.80%
20211213	2017	4.88%
20211212	2076	2.88%
20211211	2184	5.07%
20211208	2221	1.68%
20211109	2057	-7.67%
20211108	2043	-0.68%
20211107	1954	-4.45%
20211106	1864	-4.72%
20211103	1802	-3.38%
20211102	1718	-4.77%
20211101	1648	-4.16%
20211031	1574	-4.59%
20211030	1650	4.72%
20211027	1706	3.34%

20211026	1698	-0.47%
20211025	1679	-1.13%
20211023	1676	-0.18%
20211020	1758	4.78%
20211019	1780	1.24%
20211018	1706	-4.25%
20211017	1704	-0.12%
20211016	1768	3.69%
20211013	1704	-3.69%
20211012	1744	2.32%
20211011	1831	4.87%
20211010	1830	-0.05%
20211009	1863	1.79%
20211006	1944	4.26%
20211004	1932	-0.62%
20211003	1878	-2.83%
20211002	1839	-2.10%
20210929	1845	0.33%
20210928	1833	-0.65%
20210926	1862	1.57%
20210925	1885	1.23%
20210922	1814	-3.84%
20210921	1855	2.24%
20210920	1944	4.69%
20210919	1944	0.00%
20210918	2004	3.04%
20210915	2103	4.82%
20210914	2122	0.90%
20210913	2074	-2.29%
20210912	2059	-0.73%
20210911	2160	4.79%
20210908	2178	0.83%
20210907	2160	-0.83%
20210906	2251	4.13%
20210905	2276	1.10%
20210904	2253	-1.02%
20210901	2229	-1.07%
20210831	2220	-0.40%
20210830	2319	4.36%
20210829	2436	4.92%
20210828	2474	1.55%
20210825	2364	-4.55%

Appendix II – Call Option Prices from Market Transactions and Price Predictions from the Black-Scholes Model

Date of transaction	Stock Price from Market Data	Call Option Market Closing Prices	Fair Price of a Call Option using Black-Scholes
1/25/2023	3031	994	396.35
1/24/2023	3135	1059	387.08
1/23/2023	3137	1072	379.81
1/22/2023	3231	1157	369.53
1/21/2023	3284	1268	364.25
1/18/2023	3390	1405	360.39
1/17/2023	3473	1437	359.1
1/11/2023	3341	1400	351.28
1/10/2023	3486	1517	349.96
1/9/2023	3460	1507	348.65
1/8/2023	3424	1480	347.33
1/7/2023	3411	1494	346
1/4/2023	3250	1344	342.01
1/3/2023	3216	1324	340.68
1/2/2023	3074	1214	339.34
1/1/2023	2934	1021	337.99
12/31/2022	2821	955	336.64
12/28/2022	2884	1006	332.58
12/26/2022	2753	890	329.85
12/25/2022	2667	789	328.48
12/24/2022	2554	715	327.11
12/21/2022	2508	695	322.97
12/20/2022	2493	699	321.58
12/19/2022	2511	727	320.19
12/18/2022	2624	802	318.79
12/17/2022	2616	792	317.39
12/14/2022	2592	762	313.16
12/13/2022	2563	713	311.74
12/12/2022	2518	657	310.32
12/11/2022	2509	669	308.89
12/10/2022	2524	681	307.46

12/7/2022	2631	743	303.14
12/6/2022	2538	685	301.69
12/5/2022	2502	644	300.24
12/4/2022	2510	662	298.78
12/3/2022	2401	608	297.31
11/30/2022	2409	646	292.89
11/29/2022	2464	677	291.41
11/28/2022	2509	688	289.92
11/27/2022	2514	700	288.42
11/26/2022	2488	702	286.92
11/23/2022	2398	639	282.39
11/22/2022	2411	643	280.86
11/21/2022	2390	639	279.34
11/20/2022	2289	652	277.8
11/19/2022	2180	381	276.26
11/16/2022	2200	383	271.6
11/15/2022	2180	368	270.04
11/14/2022	2135	337	268.46
11/13/2022	2159	361	266.88
11/12/2022	2188	362	265.3
11/9/2022	2212	341	260.5
11/8/2022	2199	310	258.88
11/7/2022	2118	293	257.26
11/6/2022	2019	218	255.63
11/5/2022	1959	195	254
11/2/2022	1908	146	249.04
11/1/2022	1912	148	247.37
10/31/2022	1883	148	245.69
10/30/2022	1834	112	244.01
10/29/2022	1819	90	242.31
10/26/2022	1803	77	237.17
10/25/2022	1861	81	235.44
10/24/2022	1872	95	233.7
10/23/2022	1906	112	231.94
10/22/2022	1893	119	230.18
10/19/2022	1904	126	224.83

10/18/2022	1909	134	223.02
10/17/2022	1908	128	221.2
10/16/2022	1914	132	216.37
10/15/2022	1962	169	217.53
10/12/2022	1994	205	211.93
10/11/2022	1999	205	210.03
10/10/2022	1983	195	208.12
10/9/2022	1989	200	206.2
10/8/2022	2018	231	204.26
10/4/2022	1996	229	196.34
10/3/2022	1988	236	194.32
10/2/2022	1991	238	192.27
10/1/2022	1986	236	190.21
9/28/2022	2078	297	183.91
9/26/2022	1997	269	179.61
9/24/2022	1920	222	175.2
9/21/2022	1996	298	168.39
9/20/2022	1994	263	166.07
9/19/2022	2028	282	163.71
9/18/2022	2049	299	161.32
9/14/2022	2103	346	151.39
9/13/2022	2077	332	148.81
9/12/2022	1989	247	146.18
9/11/2022	1974	254	143.5
9/10/2022	1997	272	140.77
9/7/2022	2065	350	132.25
9/6/2022	2065	360	129.28
9/4/2022	2069	402	123.16
9/3/2022	2069	404	120.01
8/31/2022	2042	447	110.45
8/28/2022	2111	500	

Appendix III – Error of Black-Scholes' Fair Price Determined

Date of transaction	Call Option Market Closing Prices	Fair Price of a Call Option using Black-Scholes	Error
1/25/2023	994	396.35	597.65
1/24/2023	1059	387.08	671.92
1/23/2023	1072	379.81	692.19
1/22/2023	1157	369.53	787.47
1/21/2023	1268	364.25	903.75
1/18/2023	1405	360.39	1044.61
1/17/2023	1437	359.1	1077.9
1/11/2023	1400	351.28	1048.72
1/10/2023	1517	349.96	1167.04
1/9/2023	1507	348.65	1158.35
1/8/2023	1480	347.33	1132.67
1/7/2023	1494	346	1148
1/4/2023	1344	342.01	1001.99
1/3/2023	1324	340.68	983.32
1/2/2023	1214	339.34	874.66
1/1/2023	1021	337.99	683.01
12/31/2022	955	336.64	618.36
12/28/2022	1006	332.58	673.42
12/26/2022	890	329.85	560.15
12/25/2022	789	328.48	460.52
12/24/2022	715	327.11	387.89
12/21/2022	695	322.97	372.03
12/20/2022	699	321.58	377.42
12/19/2022	727	320.19	406.81
12/18/2022	802	318.79	483.21
12/17/2022	792	317.39	474.61
12/14/2022	762	313.16	448.84
12/13/2022	713	311.74	401.26
12/12/2022	657	310.32	346.68
12/11/2022	669	308.89	360.11
12/10/2022	681	307.46	373.54
12/7/2022	743	303.14	439.86
12/6/2022	685	301.69	383.31

12/5/2022	644	300.24	343.76
12/4/2022	662	298.78	363.22
12/3/2022	608	297.31	310.69
11/30/2022	646	292.89	353.11
11/29/2022	677	291.41	385.59
11/28/2022	688	289.92	398.08
11/27/2022	700	288.42	411.58
11/26/2022	702	286.92	415.08
11/23/2022	639	282.39	356.61
11/22/2022	643	280.86	362.14
11/21/2022	639	279.34	359.66
11/20/2022	652	277.8	374.2
11/19/2022	381	276.26	104.74
11/16/2022	383	271.6	111.4
11/15/2022	368	270.04	97.96
11/14/2022	337	268.46	68.54
11/13/2022	361	266.88	94.12
11/12/2022	362	265.3	96.7
11/9/2022	341	260.5	80.5
11/8/2022	310	258.88	51.12
11/7/2022	293	257.26	35.74
11/6/2022	218	255.63	-37.63
11/5/2022	195	254	-59
11/2/2022	146	249.04	-103.04
11/1/2022	148	247.37	-99.37
10/31/2022	148	245.69	-97.69
10/30/2022	112	244.01	-132.01
10/29/2022	90	242.31	-152.31
10/26/2022	77	237.17	-160.17
10/25/2022	81	235.44	-154.44
10/24/2022	95	233.7	-138.7
10/23/2022	112	231.94	-119.94
10/22/2022	119	230.18	-111.18
10/19/2022	126	224.83	-98.83
10/18/2022	134	223.02	-89.02
10/17/2022	128	221.2	-93.2

10/16/2022	132	216.37	-84.37
10/15/2022	169	217.53	-48.53
10/12/2022	205	211.93	-6.93
10/11/2022	205	210.03	-5.03
10/10/2022	195	208.12	-13.12
10/9/2022	200	206.2	-6.2
10/8/2022	231	204.26	26.74
10/4/2022	229	196.34	32.66
10/3/2022	236	194.32	41.68
10/2/2022	238	192.27	45.73
10/1/2022	236	190.21	45.79
9/28/2022	297	183.91	113.09
9/26/2022	269	179.61	89.39
9/24/2022	222	175.2	46.8
9/21/2022	298	168.39	129.61
9/20/2022	263	166.07	96.93
9/19/2022	282	163.71	118.29
9/18/2022	299	161.32	137.68
9/14/2022	346	151.39	194.61
9/13/2022	332	148.81	183.19
9/12/2022	247	146.18	100.82
9/11/2022	254	143.5	110.5
9/10/2022	272	140.77	131.23
9/7/2022	350	132.25	217.75
9/6/2022	360	129.28	230.72
9/4/2022	402	123.16	278.84
9/3/2022	404	120.01	283.99
8/31/2022	447	110.45	336.55